



Profile

I solve technical problems end to end: figuring out what's needed, designing the solution, sourcing parts, and building it. My background covers electronics, measurement systems, and mechatronics, with hands-on R&D experience at TNO and a track record of running my own technical business. What drives me most is turning a vague problem into a working result.

Employment

Engineer & Measurement Specialist

Feb 2025 - Present

Netherlands Organisation for Applied Scientific Research (TNO), The Hague – Ypenburg, Hybrid working

Contributing to R&D projects focused on explosives, ammunition, and protective vehicle systems. I design, build, and run high-precision measurement setups using high-speed data acquisition systems (up to 4 MHz), and developed a closure unit enabling simultaneous measurement of up to 12 signals. I also build tools for processing and reporting measurement data, and process 3D scans in SolidWorks for 3D printing. My role combines design, hands-on building, testing, and data analysis.

Founder / Owner

Jun 2024 - Jul 2025

VB Customs, Wassenaar

Founded and ran a business specializing in customizing and maintaining Vespa and Tomos mopeds: performance upgrades, lighting modifications and custom parts. Ran the company together with a business partner, selling over 100 mopeds. I was responsible for purchasing, quotations, invoicing, and bookkeeping, while my partner handled sales and inventory. Built the company from scratch, including registration, branding, and tax administration.

Electrical Technician

Jul 2024 - Jan 2025

Van Bentem Elektrotechniek, Warmond

Part-time electrical technician serving residential clients across the Zuid-Holland region, usually working in a team of two with a company van. Handled a wide range of jobs: installing and extending distribution boards, diagnosing and resolving electrical faults, and installing home wiring and lighting. On busy days we completed up to 8 urgent jobs, diagnosing each client's problem on-site and delivering a safe, finished result.

Developer

Feb 2024 - Nov 2024

Personal details

Driver's license
A, B

Website
daanvanbeelen.com

Languages

Dutch

English

Skills

- Electrical troubleshooting — installations, distribution boards, wiring harnesses
- Data acquisition & measurement — Dewetron, DTS, LTT, PicoScope
- PCB design — KiCad, Altium Designer
- SMD soldering — ESD-safe, manual & reflow
- 3D CAD & printing — SolidWorks, Bambu Lab, UltiMaker
- 3D scanning — ZEISS scanner, ZEISS INSPECT
- Software (AI-assisted) — Python, Arduino, HTML/CSS

Hobbies

- Motorcycles, boating (licensed) & waterskiing, video editing, tennis

The Field, Leiden, Hybrid working

Developed an automated water system for plants together with a fellow student, running on a Raspberry Pi, that adjusts water usage based on environmental factors. Jointly responsible for the complete system, from concept and hardware design to a working installation; IT students later joined to further improve the automation.

Test Engineer, Intern

Aug 2023 – Feb 2024

Head Electronics B.V., Katwijk aan Zee, On-site

Worked on automated test equipment (ATE) used to verify customer PCBs for reliability and correct assembly, including stress testing and in-circuit testing of traces and solder joints. Designed and hand-soldered small-scale PCBs with SMD components for these test setups, and co-developed three PCBs with the test engineering team that went into production use. Also upgraded and maintained the test setups, resolved technical faults, and gained workshop experience with CNC machining, turning, and milling.

Junior Branch Manager

Jan 2019 – Jan 2024

IJskiosk Aan De Slag, Wassenaar, On-site at company location

Started at 13 selling ice cream behind the counter and grew into junior branch manager, responsible for opening, running, and closing one of two beach kiosks, alternating between locations. Supervised up to 4 colleagues, managed inventory and ordering, and kept daily operations running smoothly on peak days.

Education

Technicus Engineering

Sep 2021 – Jul 2025

mboRijnland, Leiden

Certificates

VCA Basic (Safety, Health, and Environment Checklist for Contractors)

Jan 2022

EU Drone Licence A1/A3 (EASA)

Mar 2025

Electrostatic Discharge (ESD) Certificate

Mar 2023

Extracurricular activities

Wiring harness for a speedboat

Designed and installed a complete wiring harness in a Piranha Espada: switches, lighting, audio system, and ignition switch with battery relay.

Designed and developed a personal e-bike

Independently built an e-bike from design to completion. Responsible for selecting components, modifying electrical systems, and conducting tests to optimize performance.